Algebra II with Trigonometry Chapter 6.1

Olympiad Solutions (Gemini)

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Solutions

1. Suppose the terminal point $P(x, y)$ determined by a real number t lies in the first quadrant and satisfies $x + y = \frac{7}{5}$. Evaluate $x - y$ given that $x > y$.

By definition, a terminal point $P(x, y)$ determined by a real number t lies on the unit circle, so its coordinates satisfy:

$$x^2 + y^2 = 1$$

We can relate $x - y$ and $x + y$ through the algebraic identity:

$$(x - y)^2 + (x + y)^2 = 2(x^2 + y^2)$$

Substitute the known values $x^2 + y^2 = 1$ and $x + y = \frac{7}{5}$ into the identity:

$$(x - y)^2 + \left(\frac{7}{5}\right)^2 = 2(1)$$

$$(x - y)^2 + \frac{49}{25} = \frac{50}{25}$$

$$(x - y)^2 = \frac{1}{25}$$

Since we are given that $x > y$, the difference $x - y$ must be positive. Therefore, taking the positive square root yields:

$$x - y = \frac{1}{5}$$

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2. Let $P(x, y)$ be a terminal point on the unit circle. Determine all possible values for the reference number $\bar{t}$ associated with P if its coordinates satisfy the equation $x^3 + y^3 = 1$.

Because $P(x, y)$ lies on the unit circle, we have $x^2 + y^2 = 1$. This implies that $|x| \leq 1$ and $|y| \leq 1$. Consequently, $x^3 \leq x^2$ and $y^3 \leq y^2$.

Adding these inequalities gives:

$$x^3 + y^3 \leq x^2 + y^2$$

We are given $x^3 + y^3 = 1$, and we know $x^2 + y^2 = 1$, which forces the inequality to be an exact equality:

$$x^3 + y^3 = x^2 + y^2 \implies x^2(1 - x) + y^2(1 - y) = 0$$

Since $x \leq 1$ and $y \leq 1$, both terms $x^2(1 - x)$ and $y^2(1 - y)$ are non-negative. For their sum to be zero, both terms must be identically zero. This yields the coordinate pairs $(1, 0)$ and $(0, 1)$.

The reference number $\bar{t}$ is the shortest distance along the unit circle from P to the x-axis.

- For $P(1, 0)$, the point is on the x-axis, so $\bar{t} = 0$.
- For $P(0, 1)$, the point is on the y-axis, so $\bar{t} = \frac{\pi}{2}$.

The possible values for the reference number are 0 and $\frac{\pi}{2}$.

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3. Let x_k be the x-coordinate of the terminal point determined by the real number $t_k = \frac{k\pi}{8}$. Mentally evaluate the product $x_1 x_2 x_3 x_5 x_6 x_7$.

The x-coordinate of the terminal point is given by $x_k = \cos\left(\frac{k\pi}{8}\right)$. We wish to evaluate the product:

$$P = \cos\left(\frac{\pi}{8}\right)\cos\left(\frac{2\pi}{8}\right)\cos\left(\frac{3\pi}{8}\right)\cos\left(\frac{5\pi}{8}\right)\cos\left(\frac{6\pi}{8}\right)\cos\left(\frac{7\pi}{8}\right)$$

Using the supplementary angle identity $\cos(\pi - \theta) = -\cos\theta$, we can pair the terms from the second quadrant with those in the first:

- $\cos\left(\frac{7\pi}{8}\right) = -\cos\left(\frac{\pi}{8}\right)$
- $\cos\left(\frac{6\pi}{8}\right) = -\cos\left(\frac{2\pi}{8}\right)$
- $\cos\left(\frac{5\pi}{8}\right) = -\cos\left(\frac{3\pi}{8}\right)$

Substituting these into the product yields:

$$P = (-1)^3\left[\cos\left(\frac{\pi}{8}\right)\cos\left(\frac{2\pi}{8}\right)\cos\left(\frac{3\pi}{8}\right)\right]^2$$

Using the complementary angle identity $\cos\left(\frac{3\pi}{8}\right) = \sin\left(\frac{\pi}{8}\right)$, the inner expression becomes:

$$\cos\left(\frac{\pi}{8}\right)\sin\left(\frac{\pi}{8}\right)\cos\left(\frac{\pi}{4}\right)$$

By the double angle formula, $\cos\left(\frac{\pi}{8}\right)\sin\left(\frac{\pi}{8}\right) = \frac{1}{2}\sin\left(\frac{\pi}{4}\right)$. Thus, the inner expression evaluates to:

$$\frac{1}{2}\sin\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{4}\right) = \frac{1}{2}\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{1}{4}$$

Squaring this result and applying the negative sign gives:

$$P = -\left(\frac{1}{4}\right)^2 = -\frac{1}{16}$$

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4. A terminal point P on the unit circle is determined by the real number $t = 11$. Using estimation, determine exactly which quadrant contains P.

To find the quadrant, we must locate the position of 11 radians relative to the standard angles. A full revolution is 2π radians, so we consider the interval $[2\pi, 4\pi]$. Specifically, we want to test if 11 falls into the fourth quadrant, which corresponds to the interval $\left(2\pi + \frac{3\pi}{2}, 2\pi + 2\pi\right) = \left(\frac{7\pi}{2}, 4\pi\right)$.

We can leverage the well-known rational upper bound for pi, $\pi < \frac{22}{7}$. Multiplying both sides by $\frac{7}{2}$ yields a beautiful lower bound for 11:

$$\frac{7\pi}{2} < \frac{7}{2}\left(\frac{22}{7}\right) = 11$$

For the upper bound, we know trivially that $\pi > 3$, so:

$$4\pi > 12 > 11$$

Combining these inequalities, we have exactly:

$$\frac{7\pi}{2} < 11 < 4\pi$$

Since 11 strictly lies between $\frac{7\pi}{2}$ (the negative y-axis) and 4π (the positive x-axis), the terminal point P is in **Quadrant IV**.

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5. Let $P_1(x_1, y_1)$ be the terminal point determined by the real number $t_1 = t$. A sequence of terminal points is generated such that P_{n+1} is determined by $t_{n+1} = \frac{\pi}{2} - t_n$ if n is odd, and $t_{n+1} = \pi + t_n$ if n is even. If $P_1 = \left(\frac{3}{5}, \frac{4}{5}\right)$, determine the exact coordinates of P_{2024}.

Let us compute the first few terms of the sequence t_n:

- $t_1 = t$
- $t_2 = \frac{\pi}{2} - t_1 = \frac{\pi}{2} - t$ (since $n = 1$ is odd)
- $t_3 = \pi + t_2 = \pi + \left(\frac{\pi}{2} - t\right) = \frac{3\pi}{2} - t$ (since $n = 2$ is even)
- $t_4 = \frac{\pi}{2} - t_3 = \frac{\pi}{2} - \left(\frac{3\pi}{2} - t\right) = t - \pi$ (since $n = 3$ is odd)
- $t_5 = \pi + t_4 = \pi + (t - \pi) = t$ (since $n = 4$ is even)

Because $t_5 = t_1$, the sequence of angles (modulo 2π) is periodic with a period of 4. Therefore, the sequence of points P_n is also periodic with period 4. Since 2024 is a multiple of 4 ($2024 = 4 \times 506$), $P_{2024} = P_4$.

The angle for P_4 is $t_4 = t - \pi$, which represents a half-circle rotation from $t_1 = t$. Geometrically, this means P_4 is the antipodal reflection of P_1 through the origin. Thus, if $P_1 = (x_1, y_1)$, then $P_4 = (-x_1, -y_1)$.

$$P_{2024} = \left(-\frac{3}{5}, -\frac{4}{5}\right)$$

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6. Find the sum of the reference numbers $\bar{t}$ for all terminal points $P(x, y)$ on the unit circle whose x-coordinates satisfy the equation $16x^4 - 16x^2 + 3 = 0$.

First, factor the given quadratic in terms of x^2:

$$16x^4 - 16x^2 + 3 = 0 \implies (4x^2 - 1)(4x^2 - 3) = 0$$

This yields two possible values for x^2: 1. $x^2 = \frac{1}{4} \implies x = \pm\frac{1}{2}$ 2. $x^2 = \frac{3}{4} \implies x = \pm\frac{\sqrt{3}}{2}$

Because P is on the unit circle ($y = \pm\sqrt{1 - x^2}$), each x-coordinate corresponds to exactly two points (one in the upper half-plane, one in the lower). The reference number $\bar{t} \in [0, \frac{\pi}{2}]$ of a terminal point $P(x, y)$ is strictly determined by its horizontal distance from the origin, meaning $\cos(\bar{t}) = |x|$.

- For $x = \pm\frac{1}{2}$, we have 4 terminal points: $(\frac{1}{2}, \frac{\sqrt{3}}{2}), (\frac{1}{2}, -\frac{\sqrt{3}}{2}), (-\frac{1}{2}, \frac{\sqrt{3}}{2}), (-\frac{1}{2}, -\frac{\sqrt{3}}{2})$.

Since $\cos(\bar{t}) = \frac{1}{2}$, all 4 points share the reference number $\bar{t} = \frac{\pi}{3}$.

- For $x = \pm\frac{\sqrt{3}}{2}$, we similarly have 4 terminal points.

Since $\cos(\bar{t}) = \frac{\sqrt{3}}{2}$, all 4 points share the reference number $\bar{t} = \frac{\pi}{6}$.

Summing the reference numbers over all $4 + 4 = 8$ terminal points gives:

$$\text{Sum} = 4\left(\frac{\pi}{3}\right) + 4\left(\frac{\pi}{6}\right) = \frac{4\pi}{3} + \frac{2\pi}{3} = \frac{6\pi}{3} = 2\pi$$